

HabitFlow API Specification

Team-404

Version 1.1

Last Updated: 2026, Sept. 30

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Endpoints marked with an Authorization parameter require the header Authorization: Bearer <token> from /api/auth/login. Every authenticated request only sees data owned by that user.

II. Projects (api/projects)

Every user has exactly one Inbox project, created at signup. It holds tasks that were not added to any other project. The Inbox cannot be created, archived or deleted through the API, and isInbox is read-only.

isFavorite and isArchived are always false when a project is created. They are changed only through Favorite / Unfavorite Project (II.H) and Archive / Unarchive Project (II.I).

A. List Projects

API Endpoint	/api/projects	
method	GET	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
archived	boolean	Optional query parameter. true returns only archived projects, false only active ones. Omit for all.
<i>Lists the projects that belong to the authenticated user, ordered by sortOrder. The Inbox is always included and listed first.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
[]	array	List of project records.
{}.projectId	integer	Project ID.
{}.name	string	Project name.
{}.description	string/null	Project description.
{}.color	string/null	Project color (hex, e.g. #2563EB).
{}.isInbox	boolean	True for the user's Inbox project (read-only).
{}.isFavorite	boolean	Favorite flag.
{}.isArchived	boolean	Archived flag.
{}.sortOrder	integer	Display order.
Sample Response		

```
[{"projectId": 1, "name": "Inbox", "description": null, "color": null, "isInbox": true, "isFavorite": false, "isArchived": false, "sortOrder": 0}, {"projectId": 10, "name": "Capstone", "description": "To-Do list for Capstone (IMPORTANT)", "color": "#0000FF", "isInbox": false, "isFavorite": true, "isArchived": false, "sortOrder": 0}]
```

Fail

Status Code	401	
Response Type	json	
Response Data	Type	Description
error	string	Authentication error.

Sample Response

```
{"error": "Unauthorized"}
```

B. Get Project

API Endpoint	/api/projects/{projectId}	
method	GET	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
projectId	integer	Existing project ID (path parameter).
<i>Returns a single project owned by the authenticated user.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
projectId	integer	Project ID.
name	string	Project name.
description	string/null	Project description.
color	string/null	Project color (hex, e.g. #2563EB).
isInbox	boolean	True for the user's Inbox project (read-only).
isFavorite	boolean	Favorite flag.
isArchived	boolean	Archived flag.
sortOrder	integer	Display order.
Sample Response		
{"projectId": 10, "name": "Capstone", "description": "To-Do list for Capstone (IMPORTANT)", "color": "#0000FF", "isInbox": false, "isFavorite": true, "isArchived": false, "sortOrder": 0}		
Fail		
Status Code	404	

Response Type	json	
Response Data	Type	Description
error	string	Project not found or not accessible.
Sample Response		
{ "error": "Project not found" }		

C. Create Project

API Endpoint	/api/projects	
method	POST	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
name	string	Project name (required, max 150 chars).
description	string/null	Optional description (max 500 chars).
color	string/null	Optional color value (hex).
sortOrder	integer	Display order (default is 0).
<i>Creates a new project for the authenticated user. New projects always start with isInbox, isFavorite and isArchived set to false. Use Favorite / Unfavorite Project (II.H) and Archive / Unarchive Project (II.I) after the project is created.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
projectId	integer	Project ID.
name	string	Project name.
description	string/null	Project description.
color	string/null	Project color (hex, e.g. #2563EB).
isInbox	boolean	True for the user's Inbox project (read-only).
isFavorite	boolean	Favorite flag.
isArchived	boolean	Archived flag.
sortOrder	integer	Display order.
Sample Response		
{ "projectId": 12, "name": "Capstone", "description": "HabitFlow requirements and development", "color": "#2563EB", "isInbox": false, "isFavorite": false, "isArchived": false, "sortOrder": 1 }		
Fail		
Status Code	400	
Response Type	json	

Response Data	Type	Description
error	string	Validation error.
Sample Response		
{"error": "Project name is required"}		

D. Update Project

API Endpoint	/api/projects/{projectId}	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
projectId	integer	Existing project ID (path parameter).
name	string	Updated project name (max 150 chars).
description	string/null	Updated description (max 500 chars).
color	string/null	Updated color (hex).
<i>Updates several details of a project at once (e.g. from an Edit Project form). Fields not sent are left unchanged. Favorite, archive and sort order are changed only through their own endpoints (II.H to II.J).</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
projectId	integer	Project ID.
message	string	Success confirmation.
Sample Response		
{"projectId": 10, "message": "Project updated successfully"}		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
{"error": "Project name cannot be empty"}		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description

error	string	Project not found or not accessible.
Sample Response		
{ <code>"error": "Invalid updating project."</code> }		

E. Rename Project

API Endpoint	/api/projects/{projectId}/name	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
projectId	integer	Existing project ID (path parameter).
name	string	New project name (required, max 150 chars).
<i>Changes only the name of a project owned by the authenticated user.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
projectId	integer	Project ID.
name	string	Updated project name.
message	string	Success confirmation.
Sample Response		
{ <code>"projectId": 10, "name": "Capstone 2", "message": "Project renamed successfully"</code> }		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
{ <code>"error": "Project name is required"</code> }		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Project not found or not accessible.
Sample Response		
{ <code>"error": "Project not found"</code> }		

F. Update Project Description

API Endpoint	/api/projects/{projectId}/description	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
projectId	integer	Existing project ID (path parameter).
description	string/null	New description (max 500 chars). null clears it.
<i>Changes only the description of a project owned by the authenticated user.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
projectId	integer	Project ID.
description	string/null	Updated description.
message	string	Success confirmation.
Sample Response		
{ "projectId": 10, "description": "Final requirements and deliverables", "message": "Project description updated successfully" }		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
{ "error": "Description must be at most 500 characters" }		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Project not found or not accessible.
Sample Response		
{ "error": "Project not found" }		

G. Update Project Color

API Endpoint	/api/projects/{projectId}/color
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method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
projectId	integer	Existing project ID (path parameter).
color	string/null	New color (hex, e.g. #2563EB). null removes the color.
<i>Changes only the color of a project owned by the authenticated user.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
projectId	integer	Project ID.
color	string/null	Updated color.
message	string	Success confirmation.
Sample Response		
{ "projectId": 10, "color": "#16A34A", "message": "Project color updated successfully" }		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
{ "error": "Invalid color format" }		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Project not found or not accessible.
Sample Response		
{ "error": "Project not found" }		

H. Favorite / Unfavorite Project

API Endpoint	/api/projects/{projectId}/favorite	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.

projectId	integer	Existing project ID (path parameter).
isFavorite	boolean	true marks the project as a favorite. false removes it from favorites.
<i>Marks or unmarks a project as a favorite. This is the only endpoint that changes isFavorite.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
projectId	integer	Project ID.
isFavorite	boolean	Updated favorite flag.
message	string	Success confirmation.
Sample Response		
<pre>{"projectId": 10, "isFavorite": true, "message": "Project added to favorites"}</pre>		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
<pre>{"error": "isFavorite is required"}</pre>		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Project not found or not accessible.
Sample Response		
<pre>{"error": "Project not found"}</pre>		

I. Archive / Unarchive Project

API Endpoint	/api/projects/{projectId}/archive	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
projectId	integer	Existing project ID (path parameter).
isArchived	boolean	true moves the project to the archive. false restores it from the archive.

Archives a project or restores it from the archive. This is the only endpoint that changes isArchived. The Inbox cannot be archived.

Success

Status Code	200	
Response Type	json	
Response Data	Type	Description
projectId	integer	Project ID.
isArchived	boolean	Updated archived flag.
message	string	Success confirmation.

Sample Response

```
{"projectId": 10, "isArchived": false, "message": "Project restored from archive"}
```

Fail

Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.

Sample Response

```
{"error": "Inbox project cannot be archived"}
```

Fail

Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Project not found or not accessible.

Sample Response

```
{"error": "Project not found"}
```

J. Update Project Sort Order

API Endpoint	/api/projects/{projectId}/sort-order	
method	PUT	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
projectId	integer	Existing project ID (path parameter).
sortOrder	integer	New display order (min 0).
<i>Changes only the display position of a project in the project list.</i>		
Success		
Status Code	200	

Response Type	json	
Response Data	Type	Description
projectId	integer	Project ID.
sortOrder	integer	Updated display order.
message	string	Success confirmation.
Sample Response		
{ "projectId": 10, "sortOrder": 2, "message": "Project sort order updated successfully" }		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
{ "error": "Sort order must be 0 or greater" }		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Project not found or not accessible.
Sample Response		
{ "error": "Project not found" }		

K. Delete Project

API Endpoint	/api/projects/{projectId}	
method	DELETE	
Request Parameters	Type	Description
Authorization	string	Bearer token of the authenticated user.
projectId	integer	ID of the project to delete (path parameter).
<i>Soft deletes a project owned by the authenticated user (sets deleted_at), whether it is active or archived. Its tasks are moved to the Inbox. The Inbox project cannot be deleted.</i>		
Success		
Status Code	200	
Response Type	json	
Response Data	Type	Description
message	string	Project deletion confirmation.
movedTaskCount	integer	Number of tasks moved to the Inbox.

Sample Response		
{ "message": "Project deleted successfully", "movedTaskCount": 4 }		
Fail		
Status Code	400	
Response Type	json	
Response Data	Type	Description
error	string	Validation error.
Sample Response		
{ "error": "Inbox project cannot be deleted" }		
Fail		
Status Code	404	
Response Type	json	
Response Data	Type	Description
error	string	Project not found or not accessible.
Sample Response		
{ "error": "Project not found" }		